WORKED SOLUTION

STEP II 2000, Question 9

Mechanics • Motion with Air Resistance

STEP II 2000, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

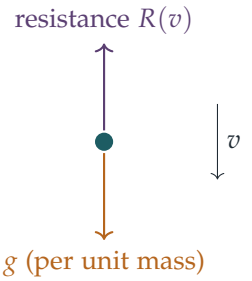
step.maths.org/questions/00-s2-q9

The key idea

Each jumper obeys Newton's second law with a resistance that depends only on speed, so we get a first-order differential equation in v and t that separates. The "speed can never reach" results come from the terminal speed: the speed at which the resistance exactly balances gravity. The solution creeps up towards it but never gets there in finite time. For Karen the quadratic factorises neatly once we write it in terms of the dimensionless speed $u = kv/g$.

Solution

We model each jumper as a particle falling vertically from rest, with constant gravitational acceleration g downwards. We take $v \geq 0$ as the downward speed and measure t from the moment of jumping. Per unit mass, the forces are gravity g downwards and the air resistance upwards, so Newton's second law gives $\dfrac{\mathrm{d}v}{\mathrm{d}t} = g - (\text{resistance per unit mass})$.



Jane's speed bound

Jane's equation of motion is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv = k\Big(\frac{g}{k} - v\Big).$$

At first $v = 0 < g/k$. While $v < g/k$ we may divide by $g/k - v > 0$ and separate the variables:

$$\int_0^v \frac{\mathrm{d}w}{g/k - w} = \int_0^t k\,\mathrm{d}s \quad\Longrightarrow\quad -\ln\Big(1 - \frac{kv}{g}\Big) = kt,$$

so

$$v = \frac{g}{k}(1 - \mathrm{e}^{-kt}).$$

This is valid for every $t \geq 0$, because the right-hand side is less than g/k for every finite t, which is the condition we used to divide. Since $\mathrm{e}^{-kt} > 0$, we have $v < g/k$ for all t. So Jane's speed tends to g/k as $t \to \infty$ but never reaches it.

Karen's speed bound

Karen's equation of motion is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv - \frac{2k^2}{g}v^2.$$

Put $u = kv/g$, a dimensionless speed. Then the right-hand side becomes

$$g(1 - u - 2u^2) = g(1 - 2u)(1 + u), \qquad \text{and} \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = \frac{k}{g}\frac{\mathrm{d}v}{\mathrm{d}t} = k(1 - 2u)(1 + u).$$

The acceleration is zero when $u = \frac{1}{2}$ (the root $u = -1$ is not a physical speed), that is, when $v = g/(2k)$. This is Karen's terminal speed.

At first $u = 0 < \frac{1}{2}$. While $0 \leq u < \frac{1}{2}$ both factors are positive, so we can separate. In partial fractions,

$$\frac{1}{(1 - 2u)(1 + u)} = \frac{2/3}{1 - 2u} + \frac{1/3}{1 + u},$$

which we can check by putting $u = -1$ and $u = \frac{1}{2}$ in $1 = A(1 + u) + B(1 - 2u)$. Hence

$$kt = \int_0^u \Big(\frac{2/3}{1 - 2w} + \frac{1/3}{1 + w}\Big)\mathrm{d}w = \frac{1}{3}\Big[-\ln(1 - 2w) + \ln(1 + w)\Big]_0^u = \frac{1}{3}\ln\frac{1 + u}{1 - 2u}.$$

Rearranging $\dfrac{1 + u}{1 - 2u} = \mathrm{e}^{3kt}$ gives

$$u = \frac{\mathrm{e}^{3kt} - 1}{2\mathrm{e}^{3kt} + 1} = \frac{1}{2} - \frac{3/2}{2\mathrm{e}^{3kt} + 1} < \frac{1}{2}.$$

This is valid for all $t \geq 0$ for the same reason as before, so Karen's speed satisfies

$$v < \frac{g}{2k} \quad \text{for all } t.$$

That is the corresponding result: her speed approaches $g/(2k)$ but never reaches it.

Jane's time to $g/(3k)$

Setting $v = g/(3k)$ in Jane's solution gives $1 - \mathrm{e}^{-kt} = \frac{1}{3}$, so $\mathrm{e}^{-kt} = \frac{2}{3}$ and

$$t = \frac{1}{k} \ln \frac{3}{2},$$

as required.

Karen's time to $g/(3k)$

> **Ask yourself** Do we need the formula for u in terms of t, or is there a quicker route?

The quicker route is the formula for t in terms of u, which we already have. Speed $g/(3k)$ means $u = \frac{1}{3}$, which is below $\frac{1}{2}$, so she does reach it. Then

$$\frac{1+u}{1-2u} = \frac{4/3}{1/3} = 4, \qquad \text{so} \qquad t = \frac{1}{3k} \ln 4 = \frac{2}{3k} \ln 2.$$

Checks

- Karen feels more resistance than Jane at every positive speed, so she should take longer to reach the same speed. Indeed $\frac{2}{3} \ln 2 \approx 0.462$ is bigger than $\ln \frac{3}{2} \approx 0.405$.
- For small t, both formulas give $v \approx gt$, which is free fall with no resistance, as it should be. Expanding $u = (\mathrm{e}^{3kt} - 1)/(2\mathrm{e}^{3kt} + 1) \approx 3kt/3 = kt$ gives $v \approx gt$.
- We stepped both equations numerically from rest (Runge–Kutta, with $g = 9.8$, $k = 0.7$). The speeds stayed below 14 and 7, which are g/k and $g/(2k)$, and the speed $g/(3k)$ was reached at $t \approx 0.5792$ and $t \approx 0.6601$. These agree with $k^{-1} \ln \frac{3}{2}$ and $\frac{2}{3k} \ln 2$ to about 10^{-9}.

Common slips

- Arguing "the speed can never reach g/k" just from the acceleration being positive. That shows the speed increases, but not that it stays below g/k. You need the explicit solution, or an argument that the time to reach the terminal speed is infinite.
- Getting the sign of the $\ln(1-2u)$ term wrong. Integrating $\frac{1}{1-2u}$ gives $-\frac{1}{2} \ln(1-2u)$.

Beyond STEP

Both equations are autonomous: $\mathrm{d}v/\mathrm{d}t = f(v)$. The terminal speed is a stable equilibrium of such an equation. A uniqueness theorem for ODEs shows that a solution can never reach an equilibrium in finite time when f is smooth, which is the general reason behind both bounds. Karen's equation is a Riccati equation, and the same

logistic-type behaviour appears in population models. Boyce and DiPrima, *Elementary Differential Equations*, covers autonomous equations and their equilibria, and Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses motion with resistance.

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